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Earth Observation Meets AI

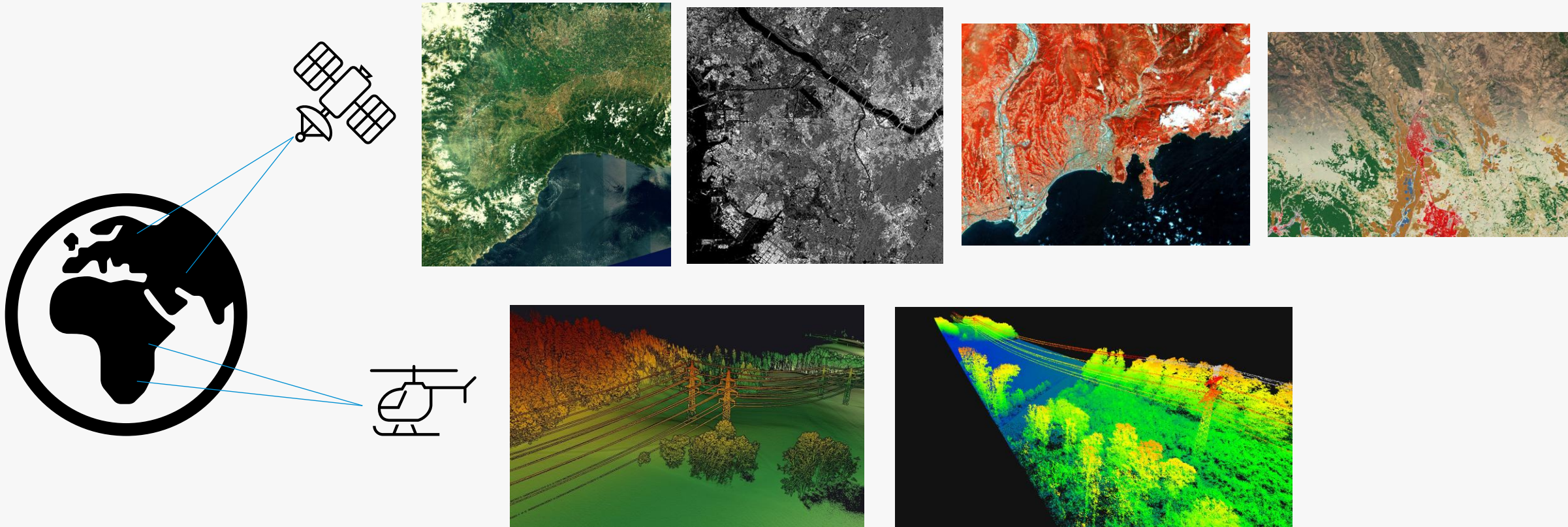
A gentle introduction to Remote Sensing

LUCA BARCO

APPLIED DATA SCIENCE PROJECT, A.Y. 2025/2026

What is remote sensing?

- Acquisition of data about the Earth from **Satellites** or **aircrafts**
- Why relevant for AI: **rich, multi-dimensional** data for prediction/classification tasks



Sources: [ESA - Sentinel-2 delivers first images](#), [Sentinel-1 – Documentation](#), [Data | Landsat Science](#), <https://www.pix4d.com/blog/lidar-photogrammetry>, [What is LiDAR? An Introduction to the Technology - Quantum Systems](#)

Key concepts in Remote Sensing

Spatial resolution

[meters per pixel]

The smallest area on the ground that a sensor can distinguish

Spectral resolution

[number and width of spectral bands]

The ability of a sensor to distinguish fine differences in wavelength.

Temporal resolution

[days]

How often a sensor revisits the same location

Radiometric resolution

[bits per pixel]

The ability of a sensor to distinguish small differences in signal intensity

Data types and formats

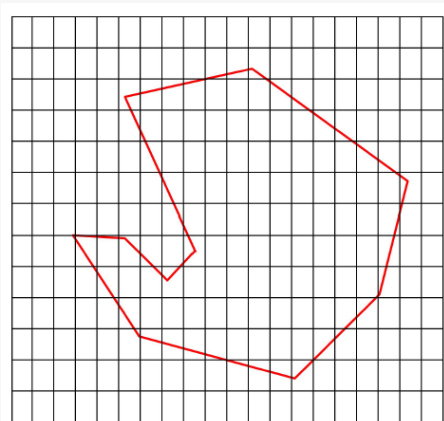
2-D

- **Raster**

Data represented as a **grid of pixels** (cells), each with a **value** representing information (e.g., reflectance, temperature, elevation).

Characteristics:

- **Continuous** data
- Easy to store **multispectral** or **temporal** information
- Examples: satellite imagery, aerial photos, DEMs
- Most common format: **GeoTiff**

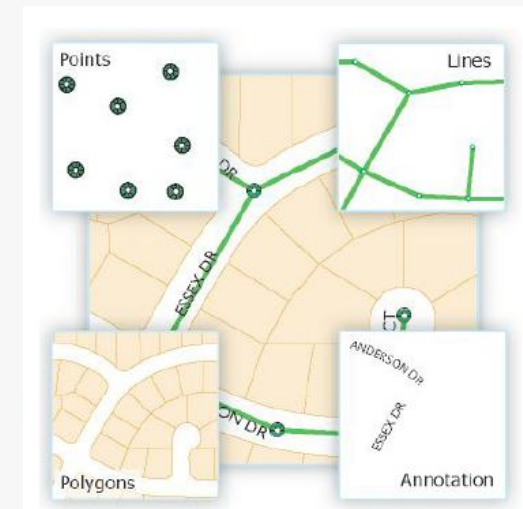


- **Vector**

Data represented by **geometric shapes**—points, lines, and polygons—that describe **discrete features**.

Characteristics:

- Represents **objects** with **clear boundaries**
- Good for roads, buildings, rivers, administrative boundaries
- Formats: **shapefiles**, **GeoJSON**



Data types and formats

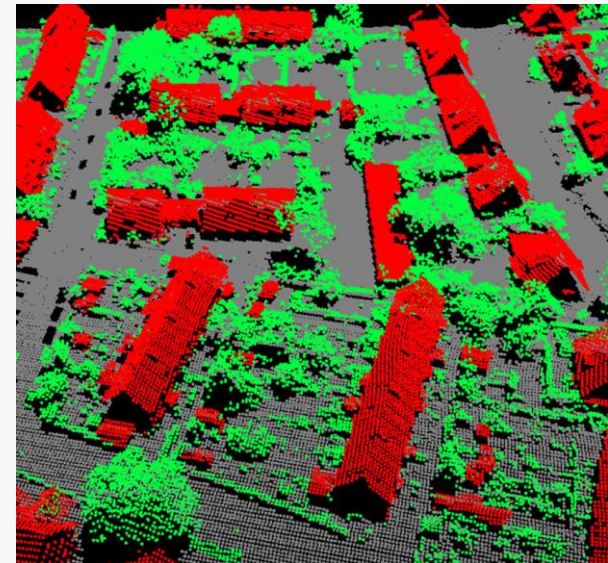
3-D

- **Point clouds**

Data represented as a **collection of points** in **3D space**, each with coordinates (X, Y, Z) and sometimes additional attributes (e.g., intensity, color).

Characteristics:

- Discrete 3D representation of surfaces or objects
- High spatial detail, often from LiDAR or photogrammetry
- Examples: LiDAR scans, 3D terrain models, urban mapping



Sensor types

different sensors provide complementary information for AI

- **Optical sensors**

- *Capture reflected sunlight in visible and near-infrared wavelengths.*
- Applications: Land cover and hazards mapping, vegetation monitoring, urban analysis.
- Examples: Sentinel-2, Landsat



- **Radar/SAR sensors**

- *Active sensors that emit microwave signals and measure their reflection from the Earth's surface.*
- Applications: Surface structure mapping, flood monitoring, soil moisture, all-weather imaging.
- Examples: Sentinel-1

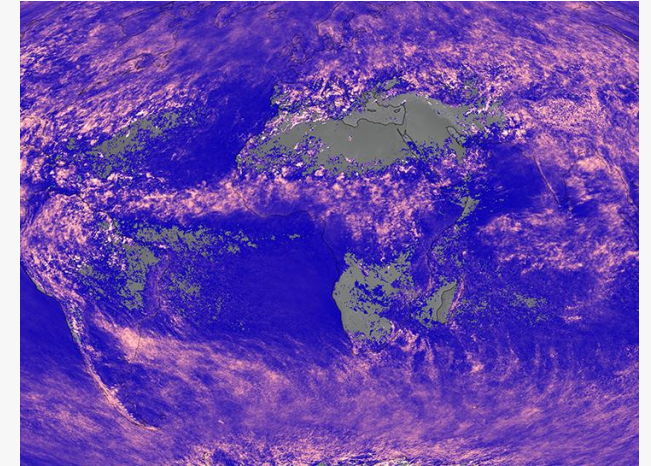


Sensor types

different sensors provide complementary information for AI

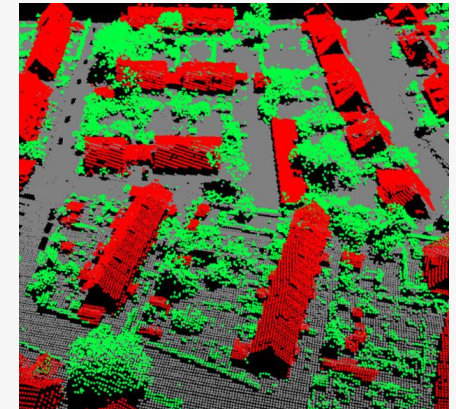
- **Thermal sensors**

- *Measure emitted thermal infrared radiation to capture surface temperature.*
- Applications: Fire detection, urban heat islands, water temperature monitoring.
- Example: MODIS, VIIRS, Sentinel 3 SLSTR



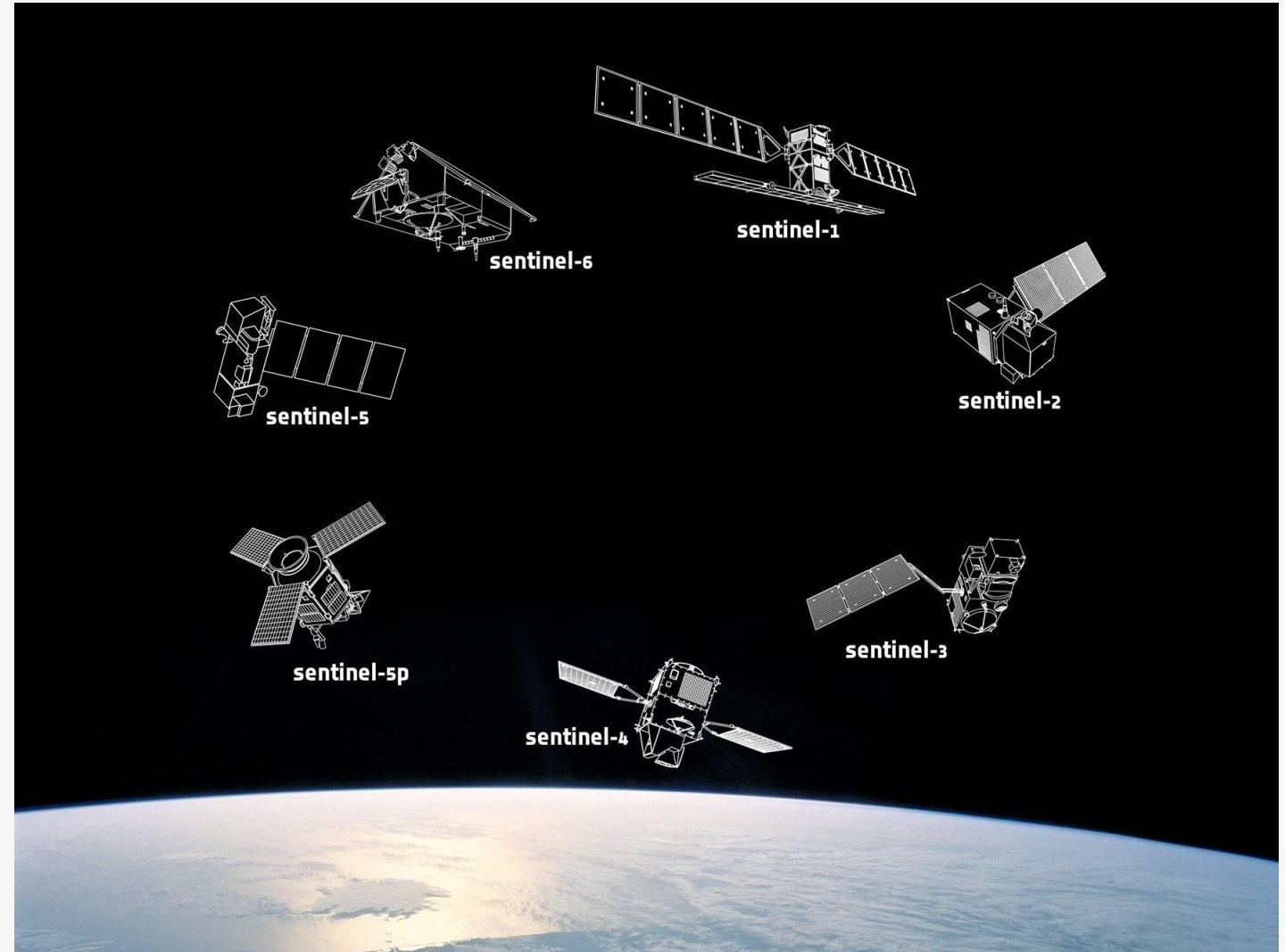
- **LiDAR (Light Detection and Ranging)**

- *Active sensor that uses laser pulses to measure distances and generate 3D point clouds.*
- Applications: High-resolution elevation models, forest canopy height, urban mapping.



The Sentinel Missions (Copernicus Programme)

- The **Copernicus Sentinel missions**, developed by ESA for the EU, provide **free and open** Earth observation data to support environmental monitoring, climate change studies, and disaster management.
- Each Sentinel satellite is designed with a specific focus—**optical, radar, ocean, or atmospheric monitoring**—ensuring continuous, long-term global coverage.



Sentinel-1

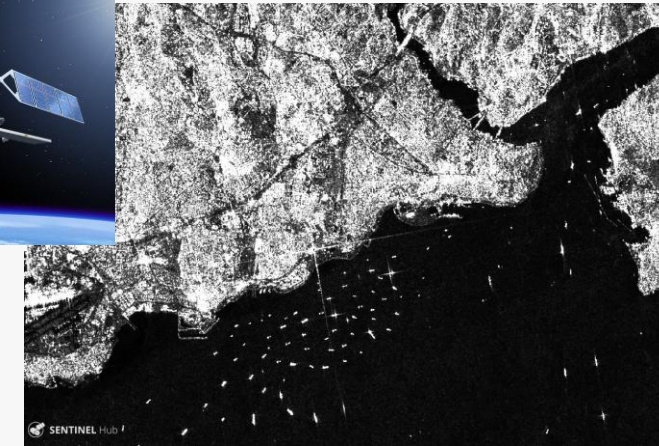
Radar (SAR) sensor

- **Sentinel-1 (C-Band Synthetic Aperture Radar SAR)**

- Sentinel 1-A launched on 2014, Sentinel 1-B launched on 2016
- All-weather, day/night imaging
- Spatial resolution: 20m/pixel
- Revisit: 6–12 days
- Usage: floods, land deformation, ships, infrastructure



[Sentinel-1](#)



Product	Definition / Processing	Characteristics	Main Uses
SLC (Single Look Complex)	Raw SAR data after focusing, with phase and amplitude preserved	Complex values (real + imaginary), slant range geometry	Interferometry (InSAR), coherence analysis, deformation studies
GRD (Ground Range Detected)	SLC data detected, multi-looked, projected to ground range	Amplitude only (real values), reduced speckle, georeferenced	Standard mapping, change detection, flood monitoring
OCN (Ocean)	Specialized GRD product processed for ocean applications	Optimized for water surface, corrected for radiometry and incidence angle	Wave analysis, wind field retrieval, ocean currents
RTC (Radiometrically Terrain Corrected)	GRD or SLC data terrain-flattened and radiometrically corrected	Backscatter normalized to flat terrain, georeferenced	Quantitative analysis on land, landslide detection, forestry, AI applications

Sentinel-2

Multispectral, optical sensor

- **Sentinel-2 (Optical)**

- Multispectral (13 bands, from Visible to SWIR)
- Sensitive to clouds
- Spatial resolution: 10–60 m
- Revisit: 5–10 days
- Usage: vegetation, agriculture, land cover



[Sentinel-2](#)



Product / Level	Definition / Processing	Characteristics	Main Uses
Level-0	Raw instrument data straight from the satellite	Unprocessed, not radiometrically or geometrically corrected	Calibration, archival purposes
Level-1B	Radiometrically corrected, still not georeferenced	Sensor-level corrections applied (e.g., sensor gains, dark current)	Calibration, scientific studies
Level-1C	Top-of-Atmosphere (TOA) reflectance, orthorectified	Corrected for sensor effects, mapped to UTM grid, 10–60 m resolution depending on band	Standard mapping, vegetation monitoring, change detection
Level-2A	Bottom-of-Atmosphere (BOA) reflectance, atmospheric corrected	Corrected for atmospheric effects (clouds, aerosols), ready for analysis	Vegetation indices (NDVI), land cover classification, time series for AI

Widely used as default optical product

Recommended for AI applications

Sentinel-3

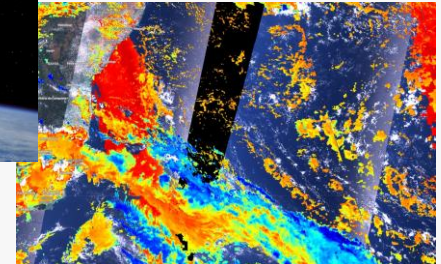
Land & Ocean Temperature Monitoring

- **Sentinel-3 (Ocean & Land Monitoring)**

- **Optical OLCI** → optical, multispectral, 300 m resolution
- **Thermal SLSTR** → thermal & optical, 500–1000 m
- **Radar altimetry SRAL** → radar altimeter, measures sea surface height
- Spatial resolution: 300m/pixel
- Revisit: 1–2 days (global coverage)
- Usage: ocean color, sea/land surface temperature, altimetry



[Sentinel-3](#)



Recommended
for AI
applications

Product / Level	Definition / Processing	Characteristics	Main Uses
Level-0	Raw instrument data directly from the satellite	Unprocessed, not calibrated or geolocated	Calibration, archival, specialized research
Level-1	Calibrated and geolocated data from each instrument	Radiances for OLCI/SLSTR, waveforms for SRAL	Scientific analysis, instrument calibration
Level-2	Geophysical variables derived from Level-1	SST (sea/land surface temperature), ocean color, altimetry parameters, chlorophyll, etc.	Environmental monitoring, climate studies, oceanography
Level-3	Gridded and mapped products, possibly merged from multiple sensors or dates	Global/regional maps, daily/weekly composites	Time series analysis, model input, AI applications

Sentinel-5P

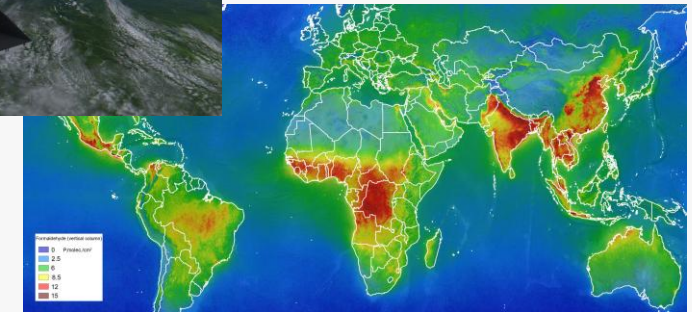
Atmosphere monitoring

- **Sentinel-5P (Atmosphere)**

- TROPOspheric Monitoring Instrument (TROPOMI), **push-broom spectrometer** in the UV–visible–SWIR range.
- Spatial resolution: 7km/pixel
- Revisit: Daily global coverage, ideal for air pollution monitoring



[ESA - Introducing Sentinel-5P](#)



Recommended for AI applications

Product / Level	Definition / Processing	Characteristics	Main Uses
Level-1B	Calibrated and geolocated radiances and spectra	Corrected for sensor effects and geolocation; contains spectral information	Calibration, scientific analysis
Level-2	Geophysical parameters derived from Level-1	Trace gas concentrations: NO ₂ , O ₃ , CO, CH ₄ , SO ₂ ; aerosol indices; UV index	Air quality monitoring, emission tracking, climate studies
Level-3	Gridded, mapped, or daily/weekly composites (optional)	Spatially and temporally averaged products, easier to use in models	Time series analysis, AI applications for environmental monitoring

Sentinel-6

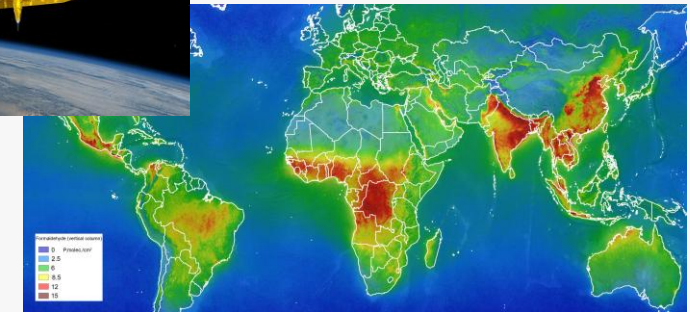
Ocean monitoring

- **Sentinel-6**

- **Radar altimeter satellite**
- Provide information on geophysical variables relating to the **ocean**:
 - **Sea-surface height (SSH)**: The distance between the satellite and the ocean surface, corrected for atmospheric and geophysical effects.
 - **Significant wave height (SWH)**: The average height (trough to crest) of the highest one-third of ocean waves.
 - **Surface wind speed**: the wind speed just above the ocean surface, derived from radar backscatter intensity.



[Overview - Sentinel Online](#)



Product / Level	Definition / Processing
Level-0	Raw telemetered data, dated and with initial geolocation.
Level-1	Level 0 data corrected for instrument and geometric effects.
Level-2	Level 1 data corrected for geophysical effects. and including derived geophysical products (Sea Surface Height Anomaly (SSHA), Significant Wave Height (SWH), and wind speed.
Level 2P	Homogenises the format of the data to match a set of consistent standards between all altimetry missions. Global and regional sea-level biases are corrected versus a reference mission, and empirical corrections are applied to remove errors in level 2 products (eg, time tag bias).

[Sentinel-6 altimetry level 1 data guide | EUMETSAT - User Portal](#)

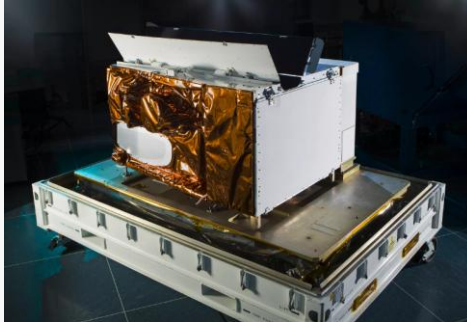
Other famous missions

Sentinel



- **Landsat**

- World's longest EO record (since 1972).
- 30 m resolution, global coverage every 16 days.
- Great for long-term change detection.



- **MODIS/VIIRS**

- Moderate resolution (250 m – 1 km).
- Daily global coverage.
- Ideal for climate, vegetation dynamics, fires.



- **EUMETSAT**

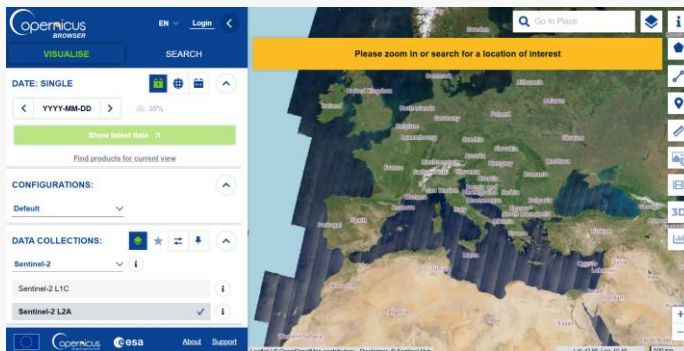
- Operates Europe's meteorological satellites.
- Geostationary (constant monitoring) + polar-orbiting.
- Key for weather, atmosphere, oceans.

How to access data

It depends on the data source

Web portals:

- [Copernicus EO Browser](#)



Python libraries:

`sentinel`, `planetary_computer`,
`rioxarray`, `stackstac`, etc.

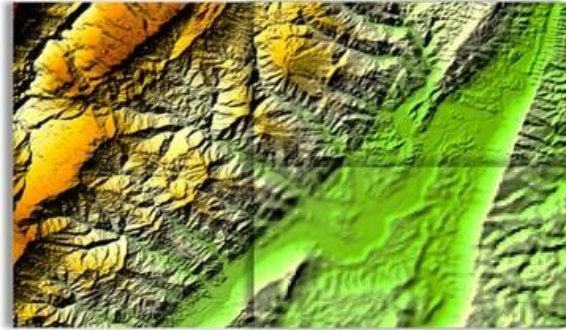
Planetary Computer, Copernicus STAC

- **STAC**: A standardized API and metadata specification to **catalog and access geospatial assets** (satellite imagery, raster/vector datasets, point clouds) in a **searchable, machine-readable way**.
- **Search by space, time, and properties** (e.g., satellite, cloud cover, sensor type).
- Returns results in JSON format, including asset URLs.
- Supports both web and programmatic access (Python clients, REST API).

EUMETSAT Data Store

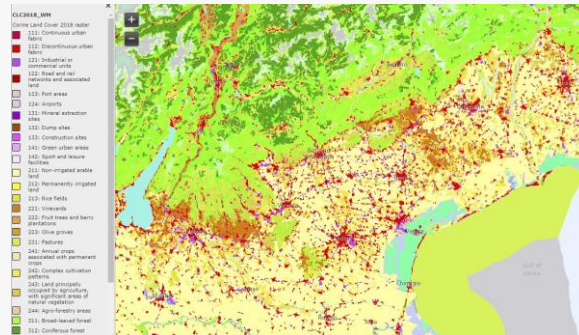
- A cloud-based platform providing access to **EUMETSAT satellite data**, including geostationary and polar-orbiting meteorological satellites.
- Supports **bulk** downloads and programmatic access via **API**.
- Data types include: imagery, atmospheric products, sea surface temperature, cloud cover, and weather parameters.
- Integrates with Python clients and Jupyter notebooks for AI and data analysis pipelines.

Useful supplementary data



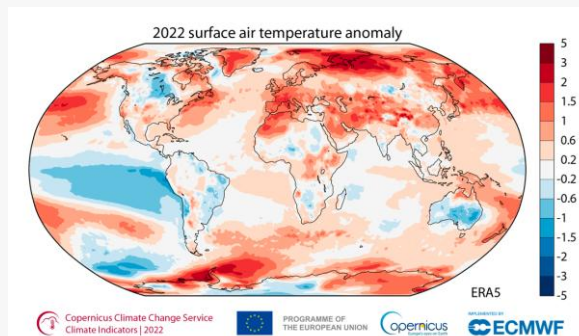
DEMs (Digital Elevation Models) - representation of the bare ground (bare earth) topographic surface of the Earth

- **SRTM (Shuttle Radar Topography Mission):** Global elevation data at ~30 m resolution; widely used for terrain analysis.
- **Copernicus DEM:** High-resolution (up to 10 m) global DEM; suitable for hydrology, flood, and slope modeling.



Land Cover - physical and biological material covering the Earth's surface

- **Copernicus EMS LC (European Environment Agency):** High-resolution land cover maps derived from satellite imagery; updated regularly
- **CORINE (Coordination of Information on the Environment):** Pan-European land cover inventory; 100 m resolution; long-term environmental monitoring.



Climate / Weather Layers (Optional)

- Temperature, precipitation, wind, etc., from sources like **ERA5**, **EUMETSAT**, or **NASA** datasets. Useful for environmental modeling and AI applications.

Main character: DATA

Remember: high quality results depend on high quality data! Choose the best data for your application



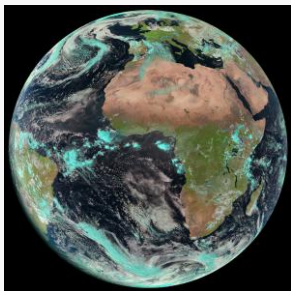
- **Sentinel-1 RTC**

- SAR (bands: VV, VH)
- Insensitive to clouds (Good for applications that need to be weather-independent)
- Max resolution: **20 m/pixel** (Not good for high-resolution application)
- Max revisit time: **12 days** (Not good for real time applications)



- **Sentinel-2 L2A**

- Optical (12 bands including RGB)
- High sensitivity to clouds (Not good for weather-dependent applications)
- Max resolution: **10 m/pixel** (Good for high resolution applications)
- Max revisit time: **10 days** (Not good for real time applications)



- **METEOSAT HRSEVIRI**

- Geostationary, (11 spectral bands including infrared, visible light, water vapour)
- Resolution: **4 km/pixel** (Not good for high-resolution application)
- Revisit time: **15 min** (Good for real time applications)

Challenges with Remote Sensing data

Spectral bands

Remote sensing images are not simply RGB but are **multispectral** or even **hyperspectral**, capturing information **far outside of the visible spectrum**.

Spatial resolution

Different sensors have **different spatial resolution** (i.e., Sentinel-1: 20m/pixel, Sentinel-2:10m/pixel), if you want to use data at different resolution you need to **resample**

Temporal dimension

Data have a **temporal dimension** (depending on the revisit time) that may be considered

Overlap

Images often have **overlap** with other images in the dataset, and need to be stitched together based on geographic metadata.

Amount

Images tend to be **very large**, so they cannot be directly passed through a neural network. They're generally **split and distributed in different files** (i.e. GeoTIFF, ESRI Shapefile, etc.) that are **read by specialized libraries** (i.e. GDAL)

Challenges with Remote Sensing data

CRS alignment

Although each image is 2D, the Earth itself is 3D. To stitch together images, they first need to be **projected onto a 2D representation of the Earth**, called a **coordinate reference system (CRS)**. Each dataset may use a different CRS, and each image within a single dataset may also be in a unique CRS. To use data from multiple layers, they must **all share a common CRS**, otherwise the data won't be properly **aligned**.

